Agro Analytics

Index

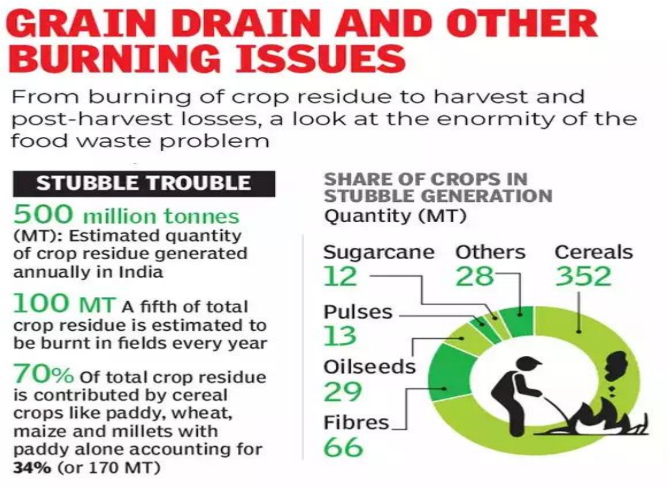
1. Project Outline
    - Problem Statement
    - Aim
2. Meet the Team
3. Our vision
4. Business Use Cases
5. Concepts Used
6. CRISP-DM Methodology
    - Business Understanding
    - Data Understanding
    - Data Preparation
    - Model Building
    - Testing and Evaluation
    - Deployment
7. Challenges faced
8. Our Learnings
9. Future Enhancement
10. Conclusion
11. Bibliography
12. Demo

Project Outline

CORPORATE
GURUKUL

Problem Statement

The Indian agricultural sector faces challenges of inefficiency and sustainability amid escalating food demand. The absence of an integrated platform hampers farmers' access to crucial information and inhibits the adoption of transformative technologies.



CORPORATE GURUKUL

Aim

To create a Smart Interactive Platform for Comprehensive Agricultural Intelligence which is imperative to revolutionize crop management, boost productivity, and elevate the overall efficiency of India's agricultural system.

CORPORATE GURUKUL

Our approach

CORPORATE
GURUKUL

We aim to achieve the following things:

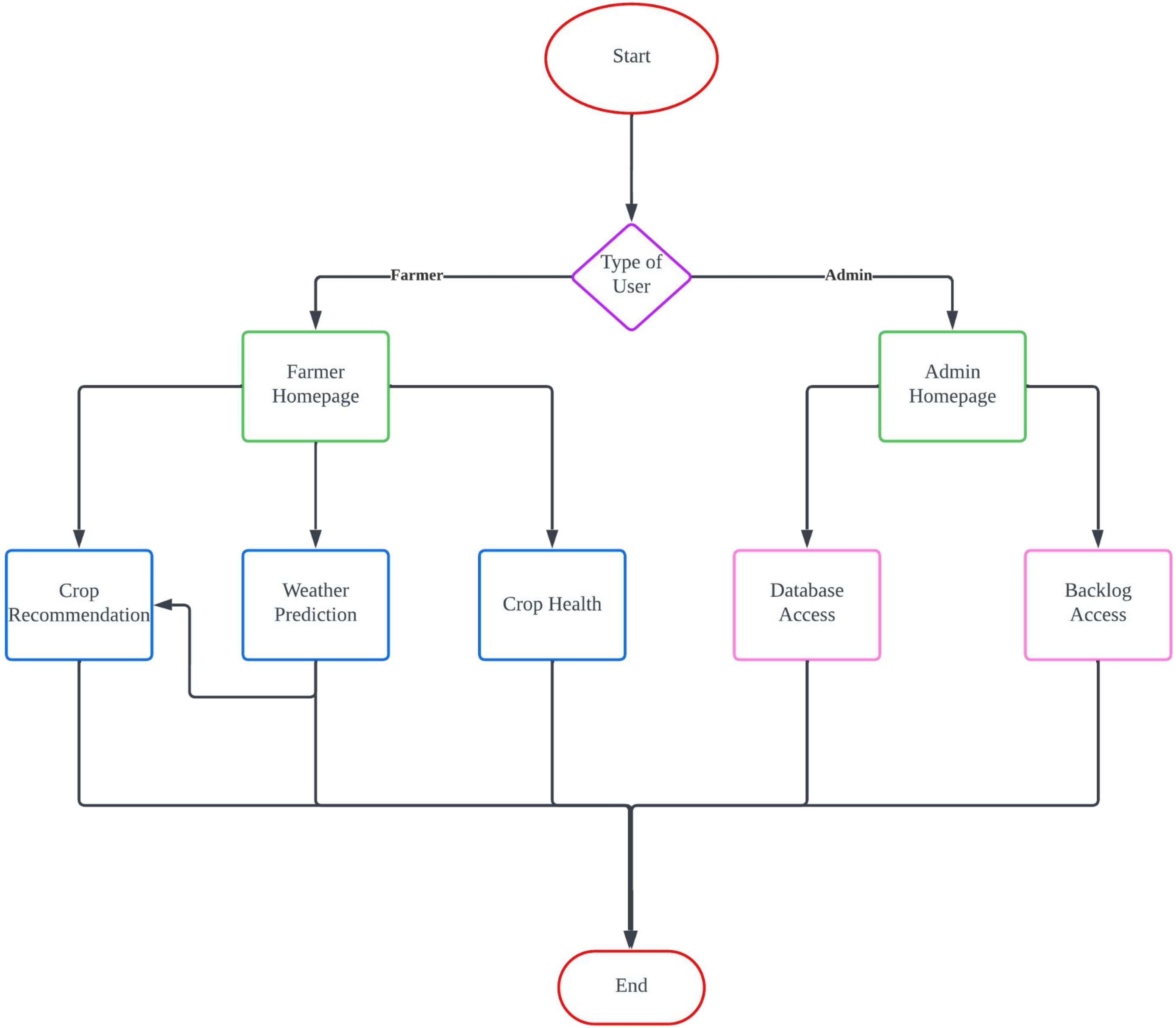
- **Precision Crop Recommendation:** Utilizing user-input temperature, humidity, rainfall, and soil pH for precise crop recommendations, aligning with dataset parameters (N, P, K, temperature, humidity, rainfall, and soil pH) for optimal farming decisions.

- **Crop Health Assessment:** Enabling users to evaluate crop health by uploading leaf images, leveraging deep learning to analyze and provide insights for proactive crop management.

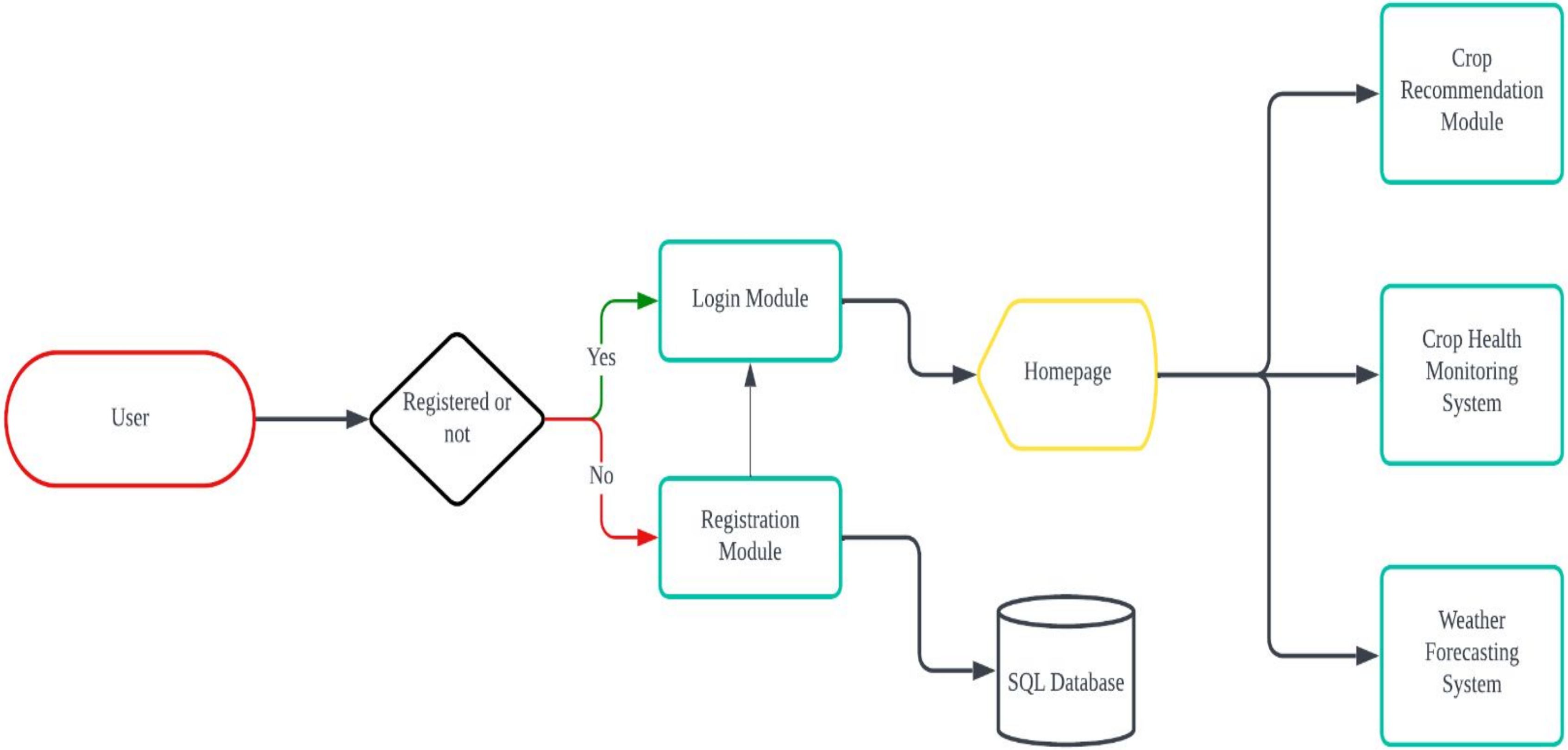
- **Long-term Weather Forecasting:** Employing a SARIMAX model trained on a comprehensive dataset spanning 25+ years to deliver accurate weather forecasts, empowering farmers with historical insights for strategic planning.

Our Vision

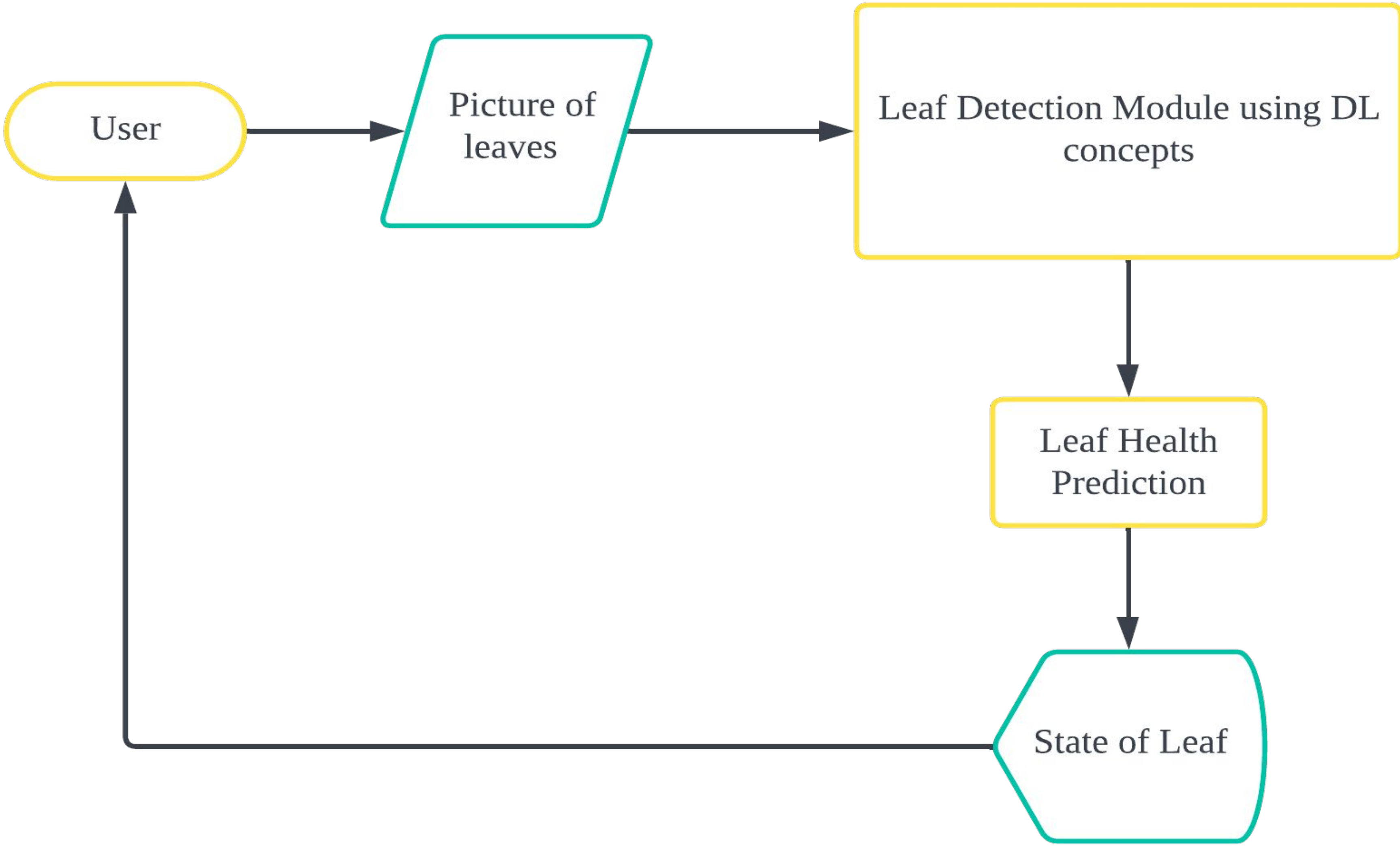
INTERFACE FLOWCHART



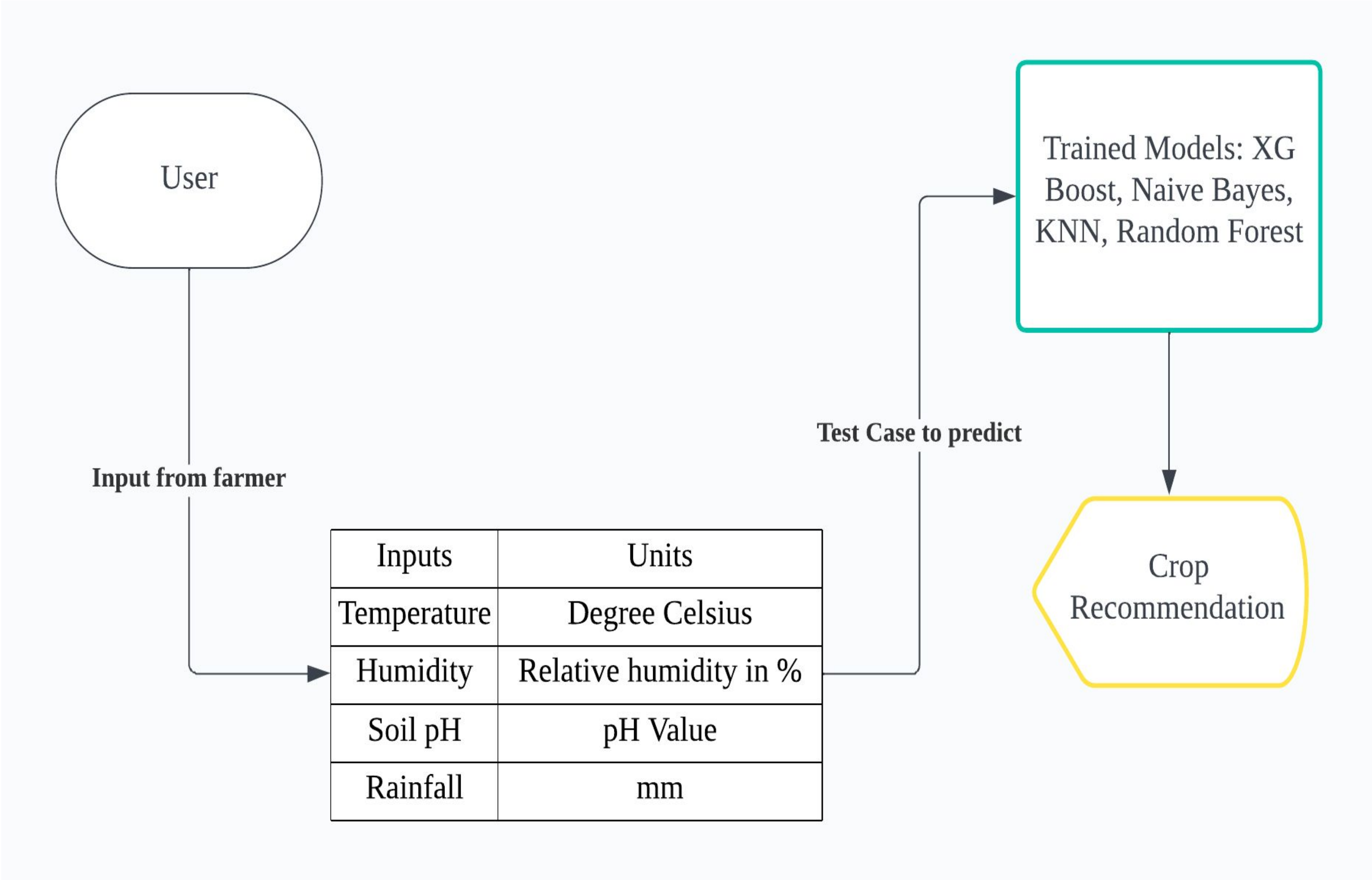
LOGIN AND MODULE REDIRECTING



CROP HEALTH ANALYSIS ARCHITECTURE

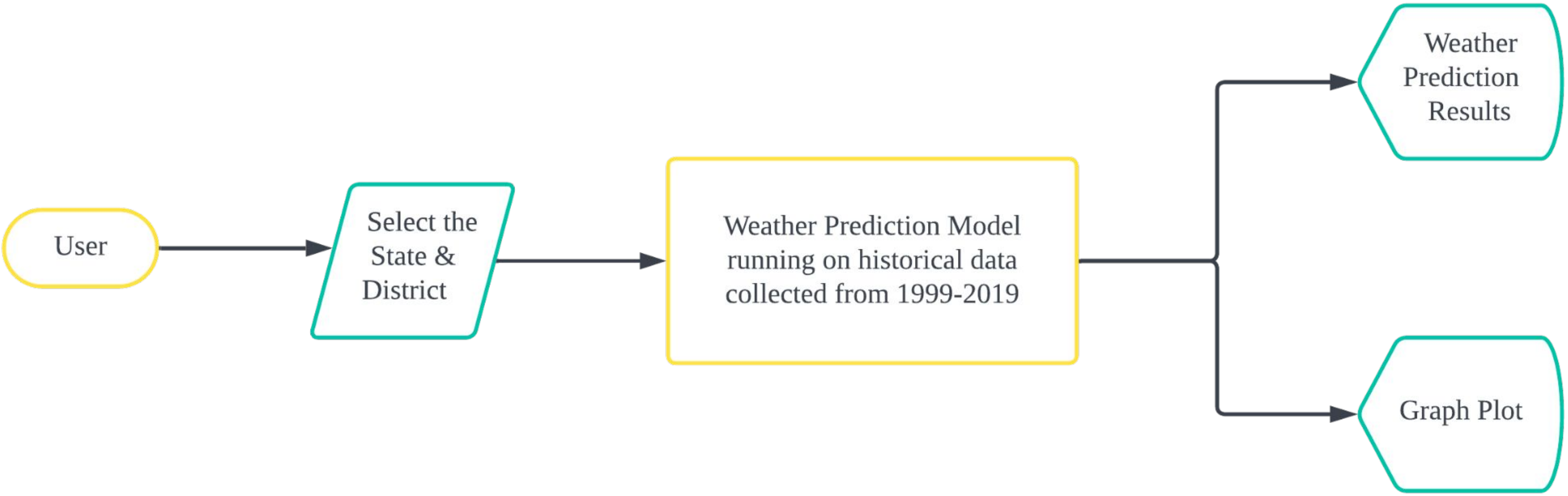


CROP RECOMMENDATION ARCHITECTURE



Inputs	Units
Temperature	Degree Celsius
Humidity	Relative humidity in %
Soil pH	pH Value
Rainfall	mm

WEATHER FORECASTING SYSTEM ARCHITECTURE



Business Use Cases

Personalised Agro-Advisory Services

- Generate revenue through tailored agro-advisory services, leveraging machine learning to provide farmers with actionable insights for optimized crop management.
- Boost productivity and promote sustainable farming.

Realtime Crop Health Analysis

- Leverages advanced sensing technologies and machine learning algorithms to continuously monitor and analyze the health status of crops,
- providing farmers with real-time insights and actionable recommendations for optimized agricultural management and improved yields.
- This innovative approach empowers farmers to make data-driven decisions, detect potential issues early, and enhance overall crop productivity.

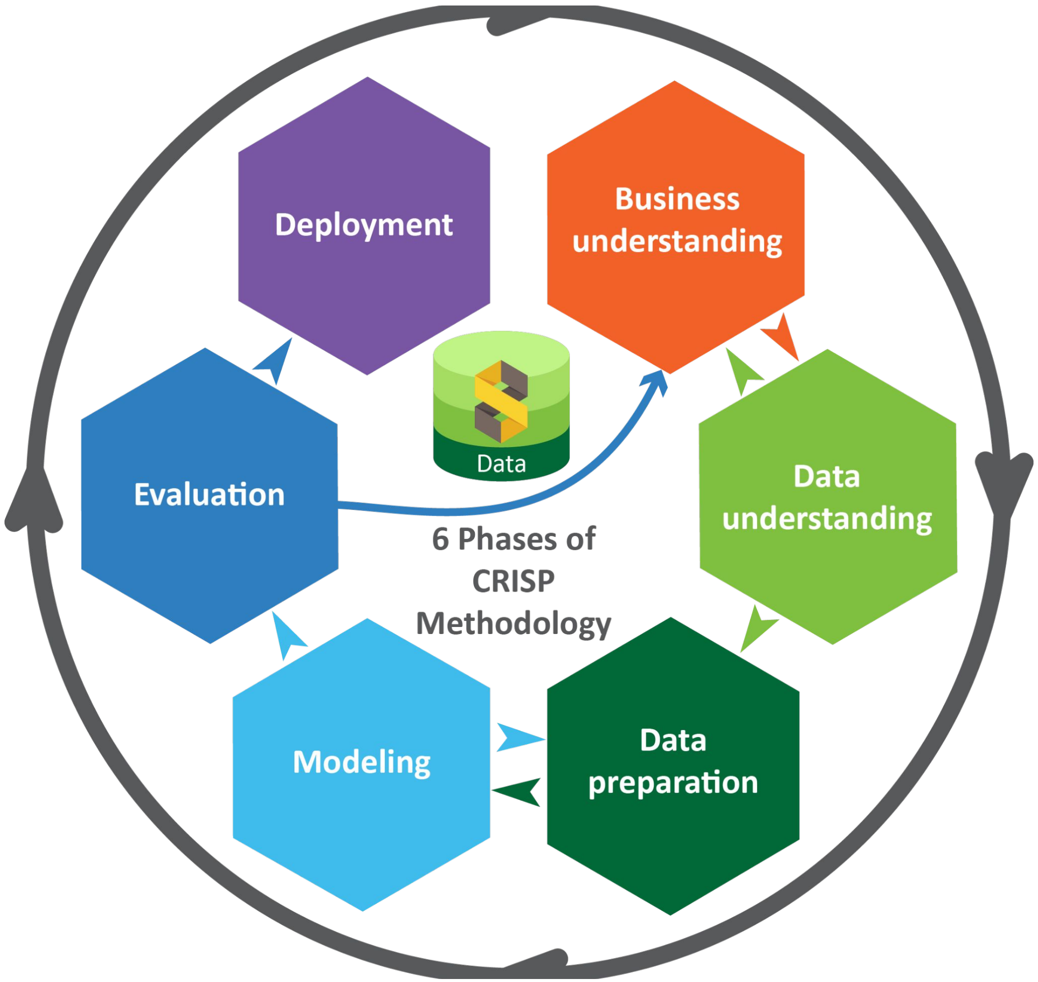
Modules Used

Crop Recommendation
1.Streamlit
2.Skicit learn library
3.XGB(eXtreme Gradient Boosting)
4.RDF(Resource Description Framework)

Crop Health Analysis
1.File Management
2.YOLO(You Only Look Once)

Weather Forecasting
1.Streamlit
2.SARIMA X(Seasonal AutoRegressive
Integrated Moving Average)

CRISP-DM Methodology



1.Business Understanding

2.Data understanding

3.Data Preparation

4.Model Building

5.Testing and Evaluation

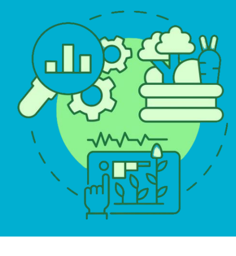
6.Deployment

Business Understanding

Business Understanding

- Precision Crop Recommendation:
    - Business Purpose: Enhance agricultural productivity and resource utilization.
    - Business Question: What crops should farmers plant based on specific environmental conditions, leading to optimal yield and resource efficiency?
- Crop Health Analysis:
    - Business Purpose: Ensure proactive crop health management.
    - Business Question: How can farmers quickly assess and address potential crop diseases or stress, minimizing losses?
- Long-term Weather Forecasting:
    - Business Purpose: Facilitate strategic planning for weather-dependent agricultural activities.
    - Business Question: How can historical weather patterns guide farmers in making informed decisions about planting, harvesting, and resource allocation?

Data Understanding



Data Understanding

- Crop Recommendation: Utilized augmented and combined datasets from various publicly available sources, focusing on essential features such as Nitrogen, Phosphorous, Potassium, pH, humidity, temperature, and rainfall for precise crop recommendations.
- Crop Health Analysis: Leveraged a dataset comprising 2,598 images across 13 plant species and 27 classes (17-10, disease-healthy) to train our model. The images, captured in natural conditions, were sourced and processed using roboflow.
- Weather Prediction: Adapted a Kaggle dataset for multiple districts in Uttar Pradesh, tailoring it to our project's needs. The dataset, spanning over 25+ years, enables accurate temperature and rainfall forecasting in the selected district chosen by the farmer.

Data Preparation

Insights into the Crop Recommendation System data

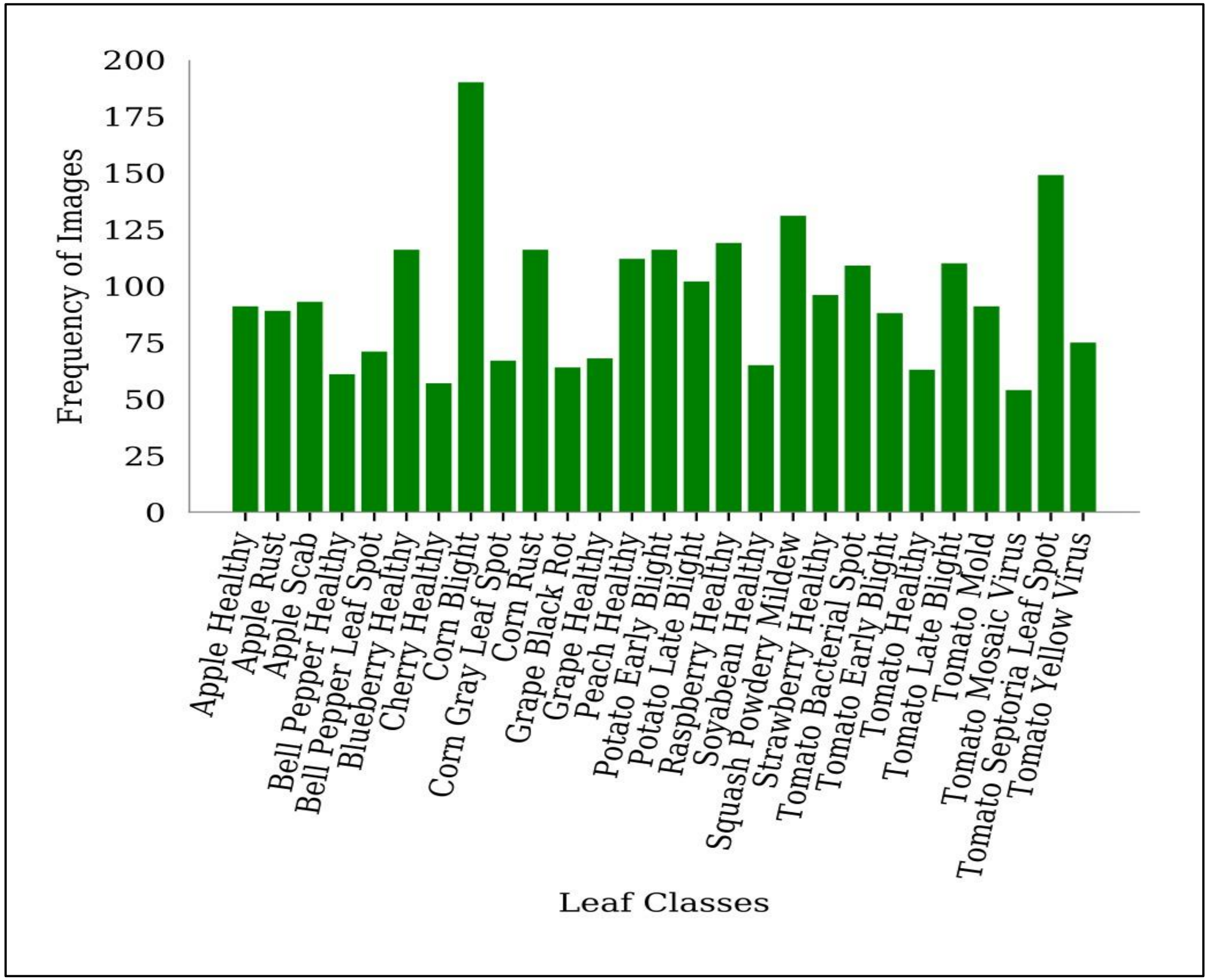
- **N (Nitrogen):** The nitrogen content in the soil, which is essential for plant growth and development.
- **P (Phosphorus):** The phosphorus content in the soil, which is crucial for energy transfer in plants.
- **K (Potassium):** The potassium content in the soil, which contributes to various physiological processes in plants.
- **Temperature:** The temperature in Celsius, which can affect the growth and development of crops.
- **Humidity:** The humidity percentage, which also influences plant growth and transpiration.
- **pH:** The pH level of the soil, indicating its acidity or alkalinity, influencing nutrient availability.
- **Rainfall:** The amount of rainfall, a critical factor for irrigation and water supply to crops.
- **Label:** The type of crop grown, serving as the target variable for classification.

Snippets for the Crop Recommendation Dataset

```
In [21]: features = df[['N', 'P','K','temperature', 'humidity', 'ph', 'rainfall']]
         target = df['label']
         labels = df['label']

In [22]: acc = []
         model = []

In [23]: from sklearn.model_selection import train_test_split
         Xtrain, Xtest, Ytrain, Ytest = train_test_split(features,target,test_size = 0.2,random_state =2)
```

	N	P	K	temperature	humidity	ph	rainfall
count	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000	2200.000000
mean	50.551818	53.362727	48.149091	25.616244	71.481779	6.469480	103.463655
std	36.917334	32.985883	50.647931	5.063749	22.263812	0.773938	54.958389
min	0.000000	5.000000	5.000000	8.825675	14.258040	3.504752	20.211267
25%	21.000000	28.000000	20.000000	22.769375	60.261953	5.971693	64.551686
50%	37.000000	51.000000	32.000000	25.598693	80.473146	6.425045	94.867624
75%	84.250000	68.000000	49.000000	28.561654	89.948771	6.923643	124.267508
max	140.000000	145.000000	205.000000	43.675493	99.981876	9.935091	298.560117

Snippets for the Crop Health Analysis Dataset



Insights into the Weather Forecasting System

STATE_UT_NAME: This column represents the name of the state or union territory.

DISTRICT: This column represents the name of the district within the state or union territory.

JAN, FEB, ..., DEC: These columns represent the monthly rainfall in millimeters for each month of the year.

ANNUAL: This column provides the total annual rainfall in millimeters for the respective district.

Jan-Feb, Mar-May, Jun-Sep, Oct-Dec: These columns represent the cumulative rainfall for specific periods of the year, combining the monthly data.

STATE_UT_NAME	DISTRICT	JAN	FEB	MAR	APR	MAY	JUN	JUL
ANDAMAN And NICOBAR ISLANDS	NICOBAR	107.3	57.9	65.2	117	358.5	295.5	285
ANDAMAN And NICOBAR ISLANDS	SOUTH ANDAMAN	43.7	26	18.6	90.5	374.4	457.2	421.3
ANDAMAN And NICOBAR ISLANDS	N & M ANDAMAN	32.7	15.9	8.6	53.4	343.6	503.3	465.4
ARUNACHAL PRADESH	LOHIT	42.2	80.8	176.4	358.5	306.4	447	660.1
ARUNACHAL PRADESH	EAST SIANG	33.3	79.5	105.9	216.5	323	738.3	990.9
ARUNACHAL PRADESH	SUBANSIRI F.D	28	48.3	85.3	101.5	140.5	228.4	217.4
ARUNACHAL PRADESH	TIRAP	42.2	72.7	141	316.9	328.7	614.7	851.9

***Disclaimer:** There are 19 columns in the dataset and we couldn't display them all.

Insights into the Weather Forecasting System (cont.)

Month: This column represents the numerical value of the month in which the temperature measurements were taken.

Day: This column represents the numerical value of the day of the month on which the temperature measurements were taken.

Year: This column represents the year in which the temperature measurements were taken.

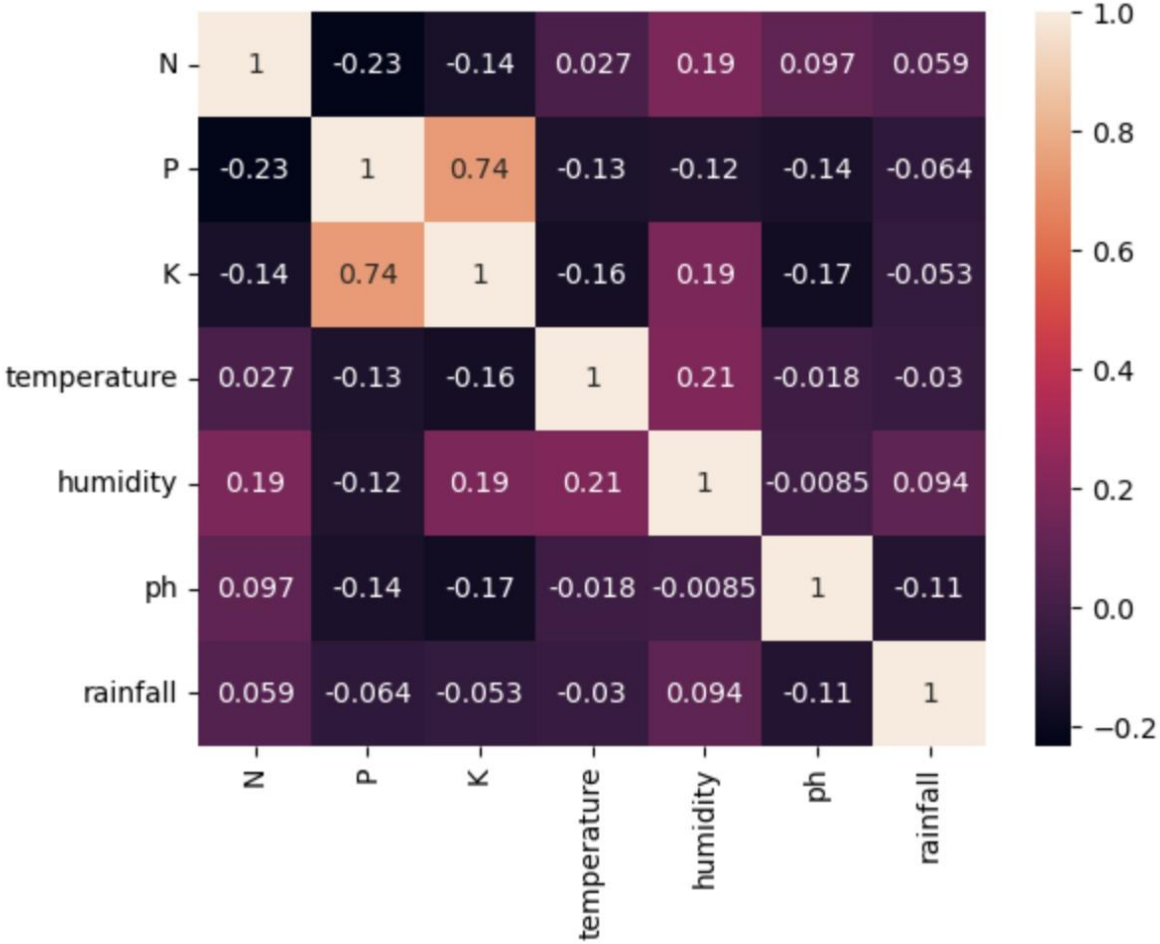
TempF: This column represents the temperature in Fahrenheit for the given day.

TempC: This column represents the temperature in Celsius for the given day.

Month	Day	Year	TempF	TempC
1	1	1999	78	25.55555556
1	2	1999	77.2	25.11111111
1	3	1999	75.9	24.38888889
1	4	1999	77.2	25.11111111
1	5	1999	74.3	23.5
1	6	1999	72.6	22.55555556
1	7	1999	70.4	21.33333333
1	8	1999	72.2	22.33333333
1	9	1999	74.4	23.55555556

Model Building

Insights into Crop Recommendation



XGBoost's Accuracy is: 0.990909090909091

	precision	recall	f1-score	support
0	1.00	1.00	1.00	13
1	1.00	1.00	1.00	17
2	1.00	1.00	1.00	16
3	1.00	1.00	1.00	21
4	1.00	1.00	1.00	21
5	0.96	1.00	0.98	22
6	1.00	1.00	1.00	20
7	1.00	1.00	1.00	18
8	0.93	0.96	0.95	28
9	1.00	1.00	1.00	14
10	0.96	1.00	0.98	23
11	1.00	1.00	1.00	21
12	1.00	1.00	1.00	26
13	1.00	0.95	0.97	19
14	1.00	1.00	1.00	24
15	1.00	1.00	1.00	23
16	1.00	1.00	1.00	29
17	1.00	1.00	1.00	19
18	1.00	1.00	1.00	18
19	1.00	1.00	1.00	17
20	1.00	0.88	0.93	16
21	1.00	1.00	1.00	15
accuracy			0.99	440
macro avg	0.99	0.99	0.99	440
weighted avg	0.99	0.99	0.99	440

RF's Accuracy is: 0.990909090909091

	precision	recall	f1-score	support
apple	1.00	1.00	1.00	13
banana	1.00	1.00	1.00	17
blackgram	0.94	1.00	0.97	16
chickpea	1.00	1.00	1.00	21
coconut	1.00	1.00	1.00	21
coffee	1.00	1.00	1.00	22
cotton	1.00	1.00	1.00	20
grapes	1.00	1.00	1.00	18
jute	0.90	1.00	0.95	28
kidneybeans	1.00	1.00	1.00	14
lentil	1.00	1.00	1.00	23
maize	1.00	1.00	1.00	21
mango	1.00	1.00	1.00	26
mothbeans	1.00	0.95	0.97	19
mungbean	1.00	1.00	1.00	24
muskmelon	1.00	1.00	1.00	23
orange	1.00	1.00	1.00	29
papaya	1.00	1.00	1.00	19
pigeonpeas	1.00	1.00	1.00	18
pomegranate	1.00	1.00	1.00	17
rice	1.00	0.81	0.90	16
watermelon	1.00	1.00	1.00	15
accuracy			0.99	440
macro avg	0.99	0.99	0.99	440
weighted avg	0.99	0.99	0.99	440

```
Naive Bayes's Accuracy is:  0.990909090909091
              precision    recall  f1-score   support

       apple       1.00      1.00      1.00        13
      banana       1.00      1.00      1.00        17
   blackgram       1.00      1.00      1.00        16
    chickpea       1.00      1.00      1.00        21
     coconut       1.00      1.00      1.00        21
      coffee       1.00      1.00      1.00        22
      cotton       1.00      1.00      1.00        20
      grapes       1.00      1.00      1.00        18
        jute       0.88      1.00      0.93        28
 kidneybeans       1.00      1.00      1.00        14
      lentil       1.00      1.00      1.00        23
       maize       1.00      1.00      1.00        21
       mango       1.00      1.00      1.00        26
   mothbeans       1.00      1.00      1.00        19
    mungbean       1.00      1.00      1.00        24
   muskmelon       1.00      1.00      1.00        23
      orange       1.00      1.00      1.00        29
      papaya       1.00      1.00      1.00        19
  pigeonpeas       1.00      1.00      1.00        18
 pomegranate       1.00      1.00      1.00        17
        rice       1.00      0.75      0.86        16
  watermelon       1.00      1.00      1.00        15

    accuracy                           0.99       440
   macro avg       0.99      0.99      0.99       440
weighted avg       0.99      0.99      0.99       440
```

Insights into Crop Health Analytics System

A glimpse of the custom data using bounty boxes from roboflow

CORPORATE GURUKUL

Snippets Of YOLO V8

```python
from ultralytics import YOLO

model = YOLO("yolov8n.yaml")  # build a new model from scratch
model = YOLO("yolov8n.pt")  # load a pretrained model (recommended for training)

# Use the model
model.train(data="/content/data.yaml", epochs=50)
```

Snippets evaluation of the model's performance on validation set

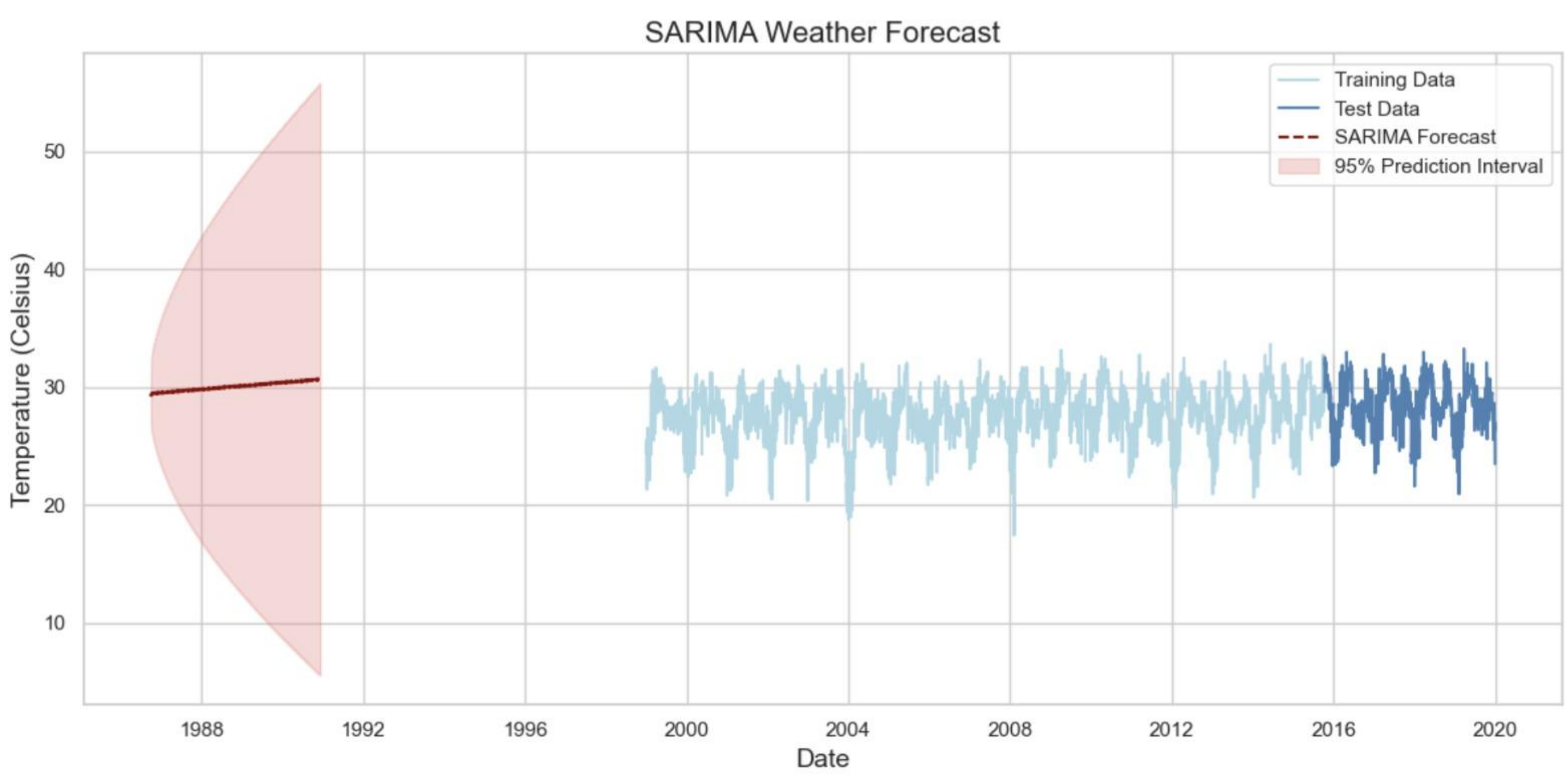
```python
# Evaluating the model's performance on the validation set
results = model.val()
```

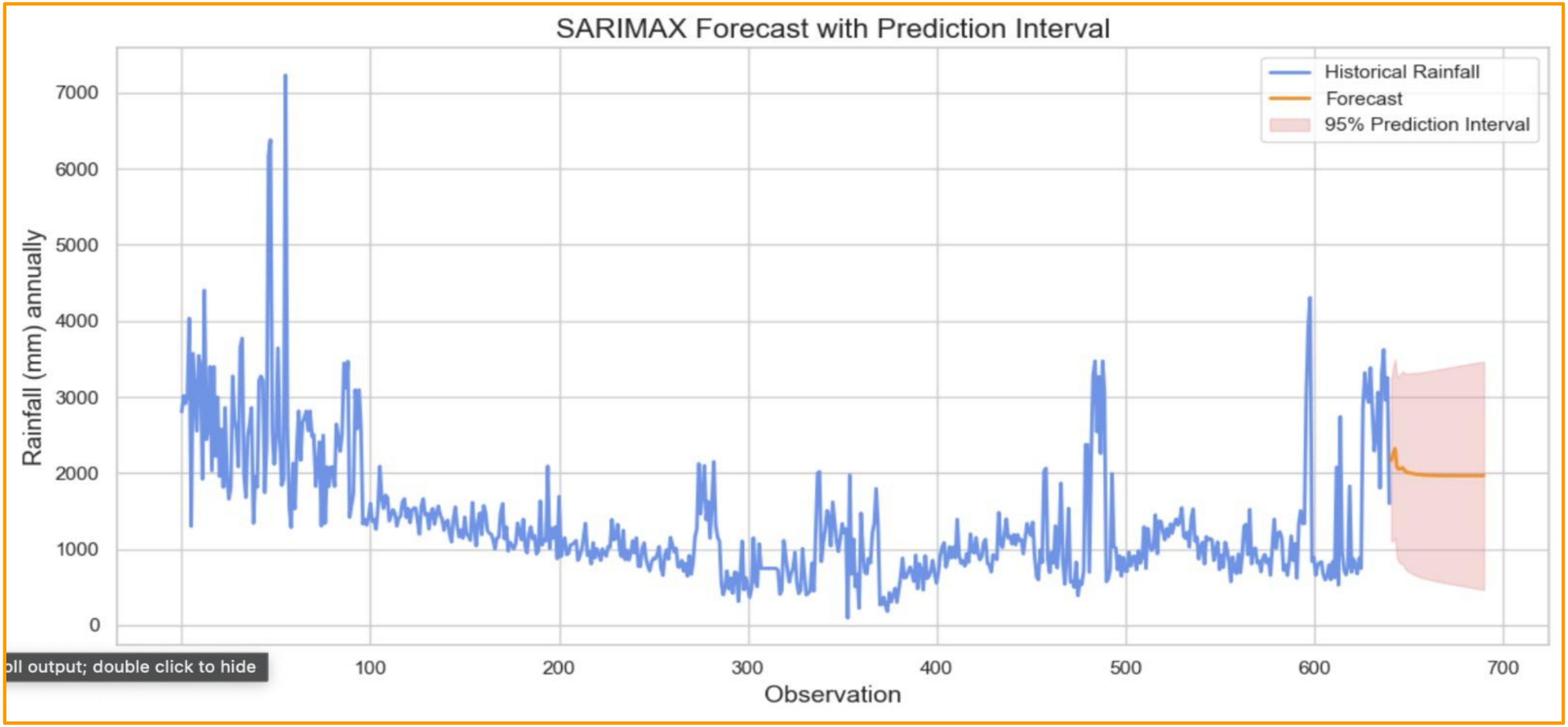
```
Ultralytics YOLOv8.0.231 🚀 Python-3.10.12 torch-2.1.0+cu121 CUDA:0 (Tesla T4, 15102MiB)
Model summary (fused): 168 layers, 3011498 parameters, 0 gradients, 8.1 GFLOPs
val: Scanning /content/valid/labels.cache... 314 images, 1 backgrounds, 0 corrupt: 100%|
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100%
                   all        314       1191      0.457      0.528      0.534      0.379
       Apple Scab Leaf        314          2      0.309        0.5      0.308      0.265
            Apple leaf        314         61      0.661       0.77      0.776      0.518
       Apple rust leaf        314          4      0.571        0.5      0.695       0.58
  Bell_pepper leaf spot       314         11      0.314      0.636      0.502      0.363
      Bell_pepper leaf        314          1          0          0     0.0169     0.0152
        Blueberry leaf        314         41      0.422      0.569      0.584      0.432
           Cherry leaf        314          4     0.0683       0.25     0.0984     0.0852
      Corn leaf blight        314         12      0.609      0.908      0.895      0.565
        Corn rust leaf        314         16          1      0.642      0.954       0.78
            Peach leaf        314        130       0.77      0.892      0.895      0.606
```

CORPORATE GURUKUL

Insights into the Weather Forecasting Model

Mean Absolute Error: 2.0310008411782325





CORPORATE
GURUKUL

A dummy's guide to flunking in ML/DL

What's the worst that can happen?

CORPORATE GURUKUL

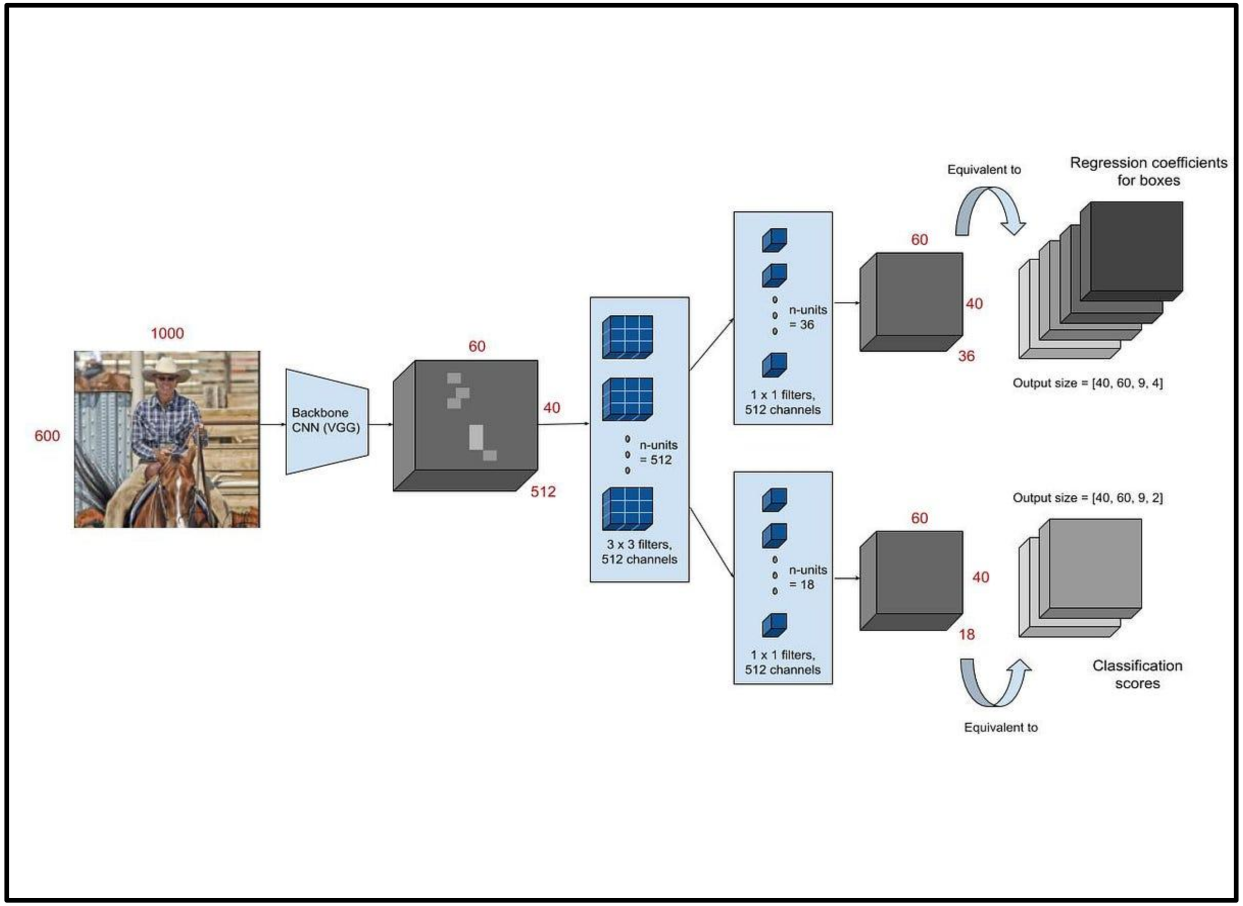
SVM, Logistic Regression, KNN, Decision Trees: What went wrong?

```
SVM's Accuracy is:  0.10681818181818181

Logistic Regression's Accuracy is:  0.9522727272727273

DecisionTrees's Accuracy is:  90.0

KNN Classifier Accuracy is: 0.97
```

CORPORATE
GURUKUL

Faster R-CNN : What went wrong?

Model fails to produce accurate results due to background noise, images with leaf from multiple classes in a dataset and low-resolution leaf images.



CORPORATE GURUKUL

How are we choosing our model ?

(warning: prioritize thought before action :/)

For Crop Recommendation System-

- ➔ Comparing the accuracy for 8 models and shortlisting the top 3.

For Crop Health Analysis System-

- ➔ Comparing accuracy of Faster R-CNN and YOLOv8 after 50 epochs.
- ➔ Faster R-CNN Model fails to produce accurate results due to background noise, images with leaf from multiple classes in a dataset and low- resolution leaf images. YOLOv8 is the latest version of YOLO by Ultralytics.
- ➔ As a cutting-edge, state-of-the-art (SOTA) model, YOLOv8 builds on the success of previous versions, introducing new features and improvements for enhanced performance, flexibility, and efficiency. YOLOv8 supports a full range of vision AI tasks, including detection, segmentation, pose estimation, tracking, and classification.
- ➔ This versatility allows users to leverage YOLOv8's capabilities across diverse applications and domains. Hence, going with YOLOv8.

For Weather Forecasting-

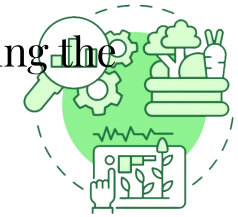
- ➔ Compared the accuracy of a stimulated Neural Network and Sarimax - ended up choosing Sarimax for its higher accuracy and efficiency while handling large datasets.

Model Comparison and Selection

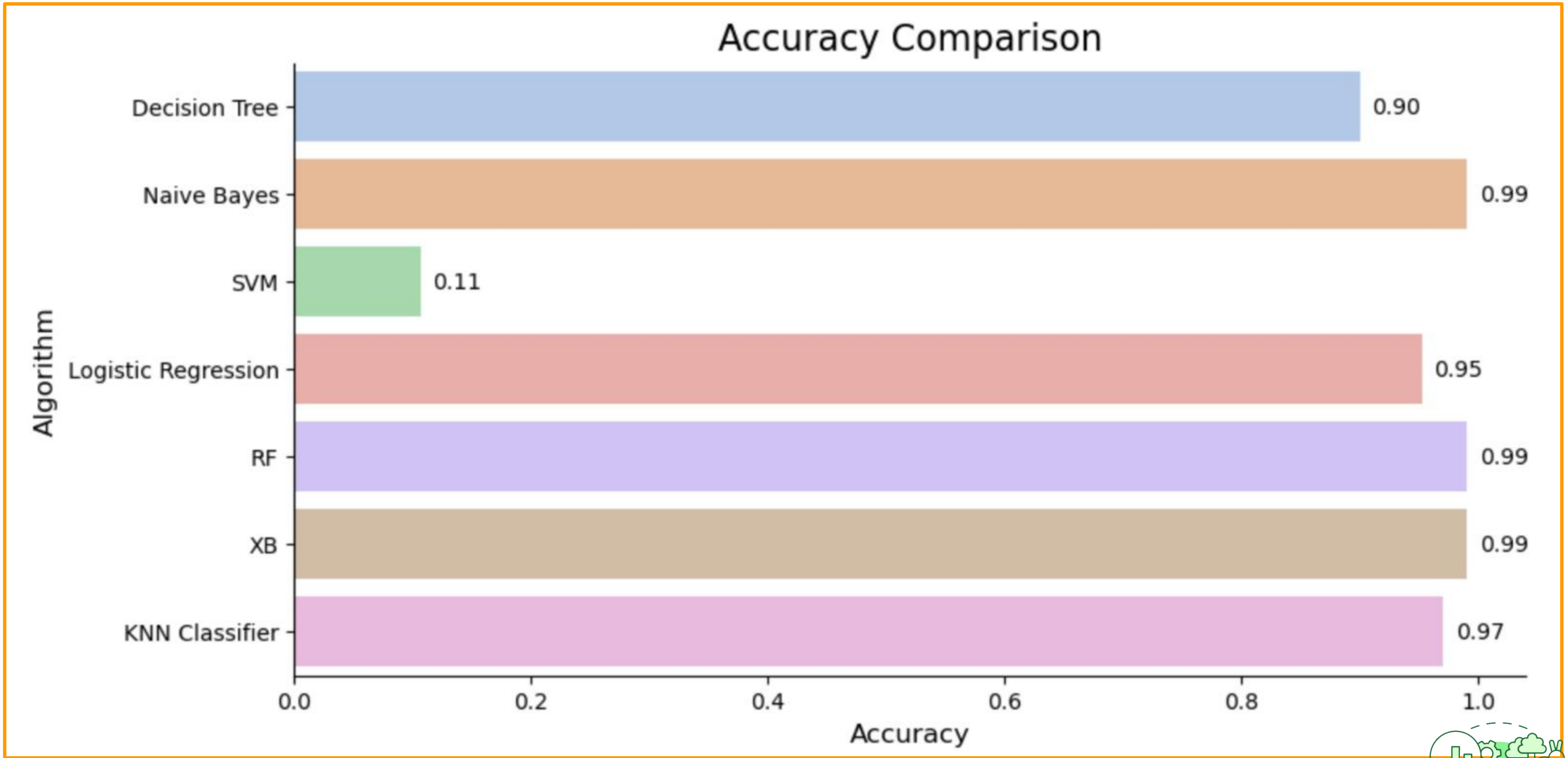
CORPORATE GURUKUL

ML Algorithms used for Crop Recommendation

Algorithms Chosen	Why?
1) **Naive Bayes (99% accuracy)**	**Efficiency and Speed:** Naive Bayes is known for its simplicity and computational efficiency. It is particularly effective for text classification and tasks with high dimensionality. **Handling Categorical Data:** Naive Bayes works well with categorical features and is robust to irrelevant features.
2) **Random Forest (99% accuracy)**	**Ensemble Learning:** Random Forest is an ensemble method that builds multiple decision trees and combines their predictions. This often leads to improved accuracy and generalization. **Robustness:** Random Forest is less prone to overfitting compared to a single decision tree. It can handle noisy and unstructured datasets effectively.
3) **XGBoost (99% accuracy)**	**High Predictive Power:** XGBoost is a powerful gradient boosting algorithm that often performs well in a wide range of tasks. It is known for its high predictive accuracy and efficiency. **Handling Missing Data:** XGBoost can handle missing values well, reducing the need for extensive data preprocessing.



CORPORATE GURUKUL



CORPORATE
GURUKUL

DL & CV Model Used For Crop Health Analysis

Model Chosen

YOLO (You Only Look Once) V8

Why?

YOLOv8 is the latest version of YOLO by Ultralytics. As a cutting-edge, state-of-the-art (SOTA) model, YOLOv8 builds on the success of previous versions, introducing new features and improvements for enhanced performance, flexibility, and efficiency. YOLOv8 supports a full range of vision AI tasks, including detection, segmentation, pose estimation, tracking, and classification.

This versatility allows users to leverage YOLOv8's capabilities across diverse applications and domains.

CORPORATE GURUKUL

Forecasting Model used Weather Prediction

Model chosen

Seasonal AutoRegressive Integrated Moving Average With Exogenous Factors (SARIMAX)

Why?

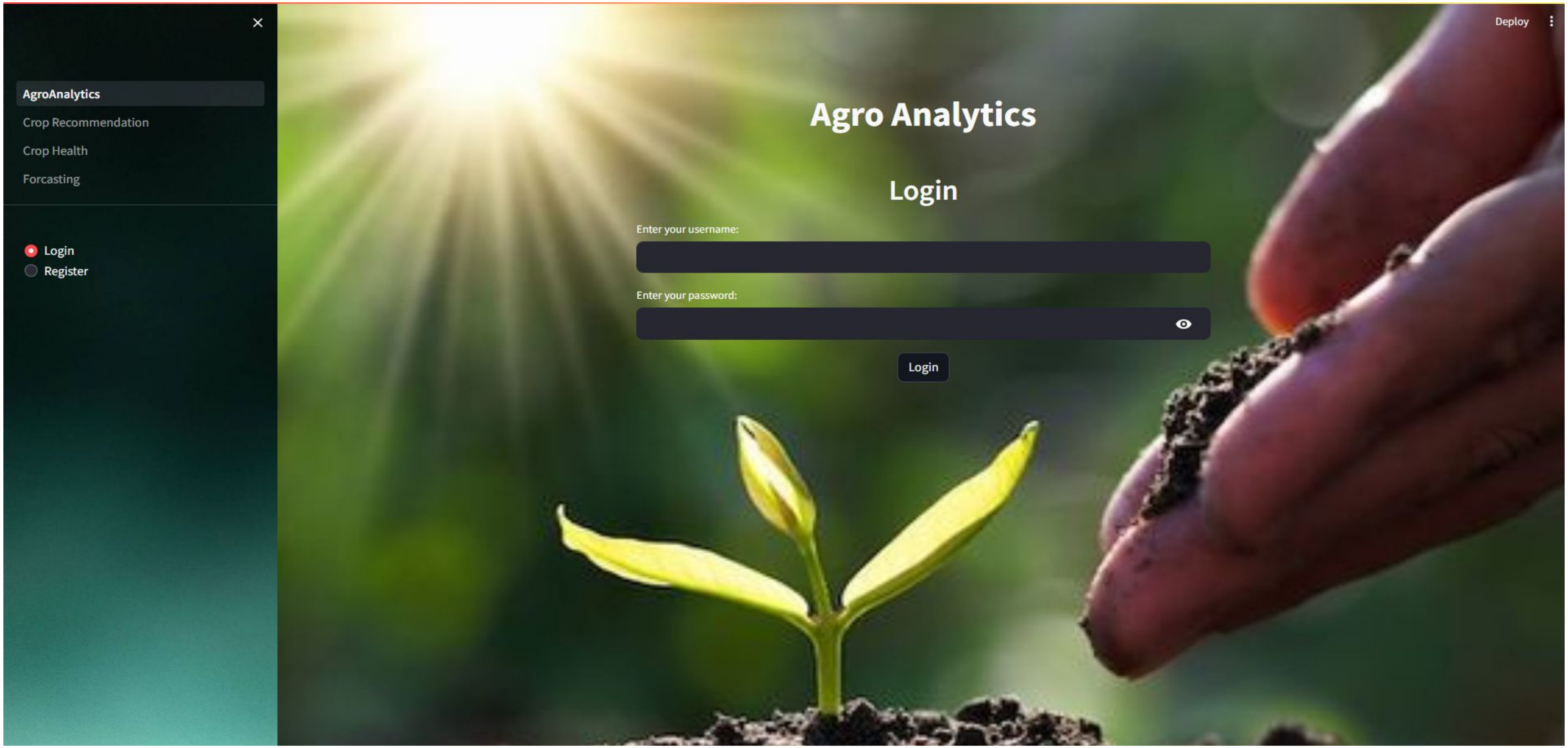
SARIMAX is advantageous due to its ability to handle both temporal patterns and external influences, providing a more comprehensive modelling approach.

It is suitable for data that exhibit trend, seasonality, and potential relationships with external factors. The model parameters and the inclusion of exogenous variables are determined based on the analysis of the time series data.

CORPORATE GURUKUL

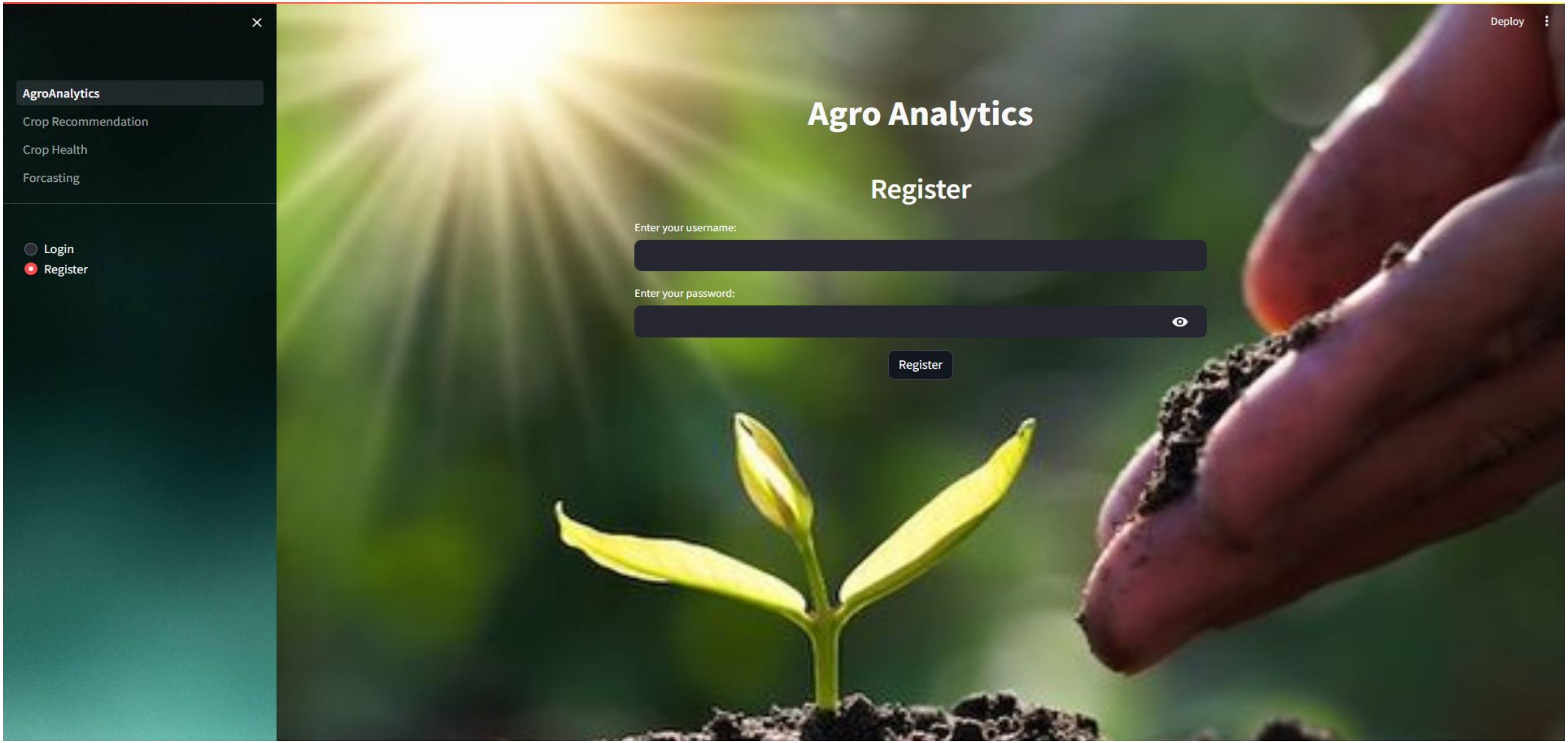
Deployment

Login Page



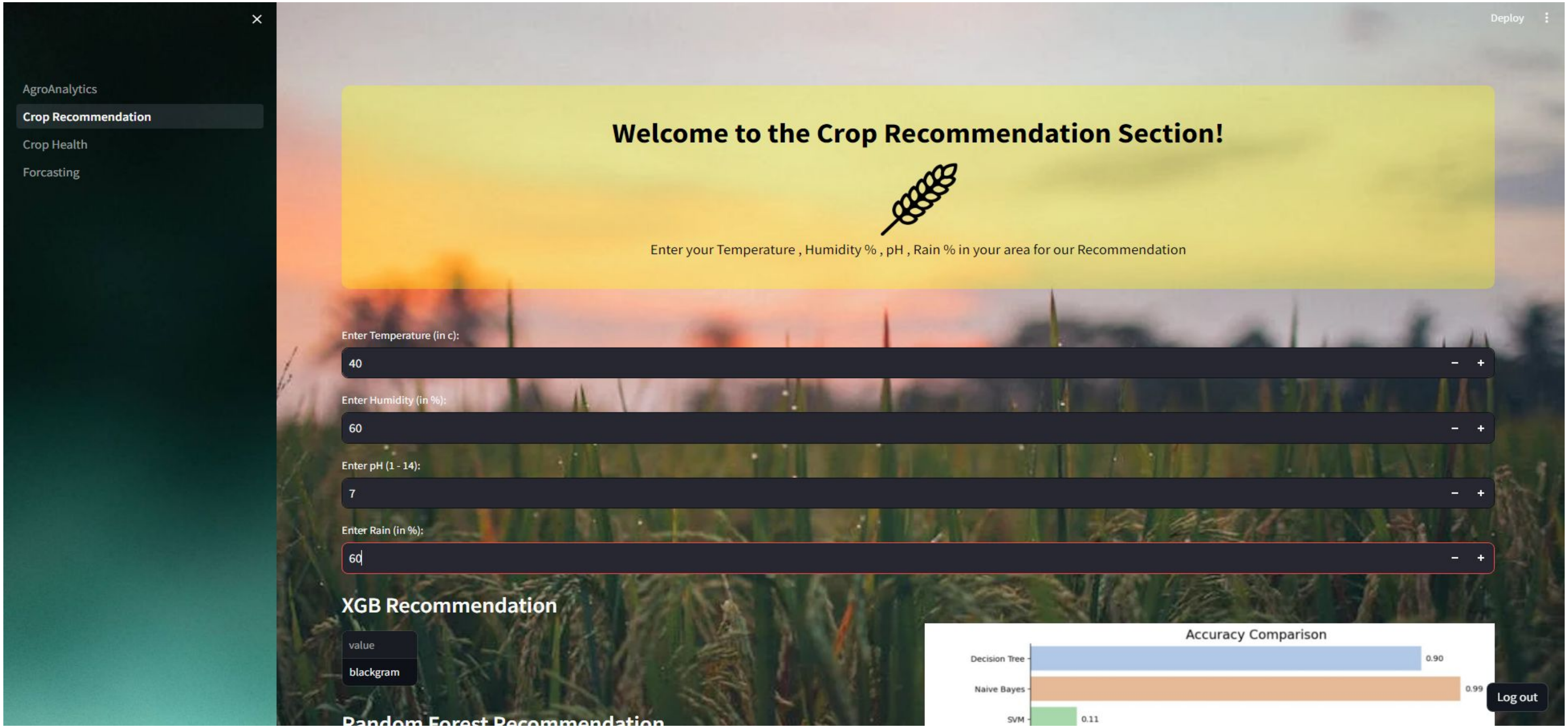
AgroAnalytics
Crop Recommendation
Crop Health
Forcasting
Login
Register
Deploy
Agro Analytics
Login
Enter your username:
Enter your password:
Login

Registration Page



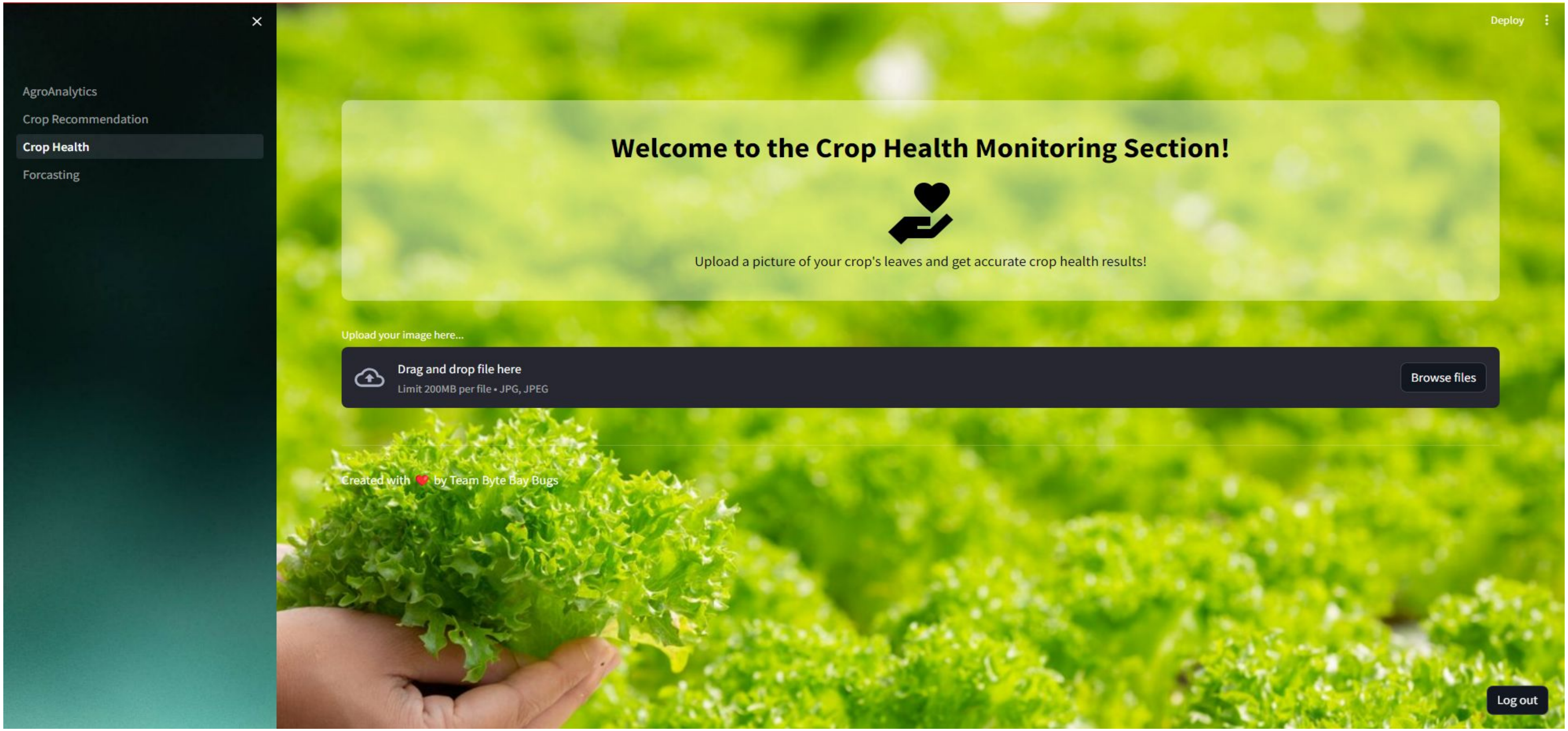
CORPORATE GURUKUL

Crop Recommendation Page

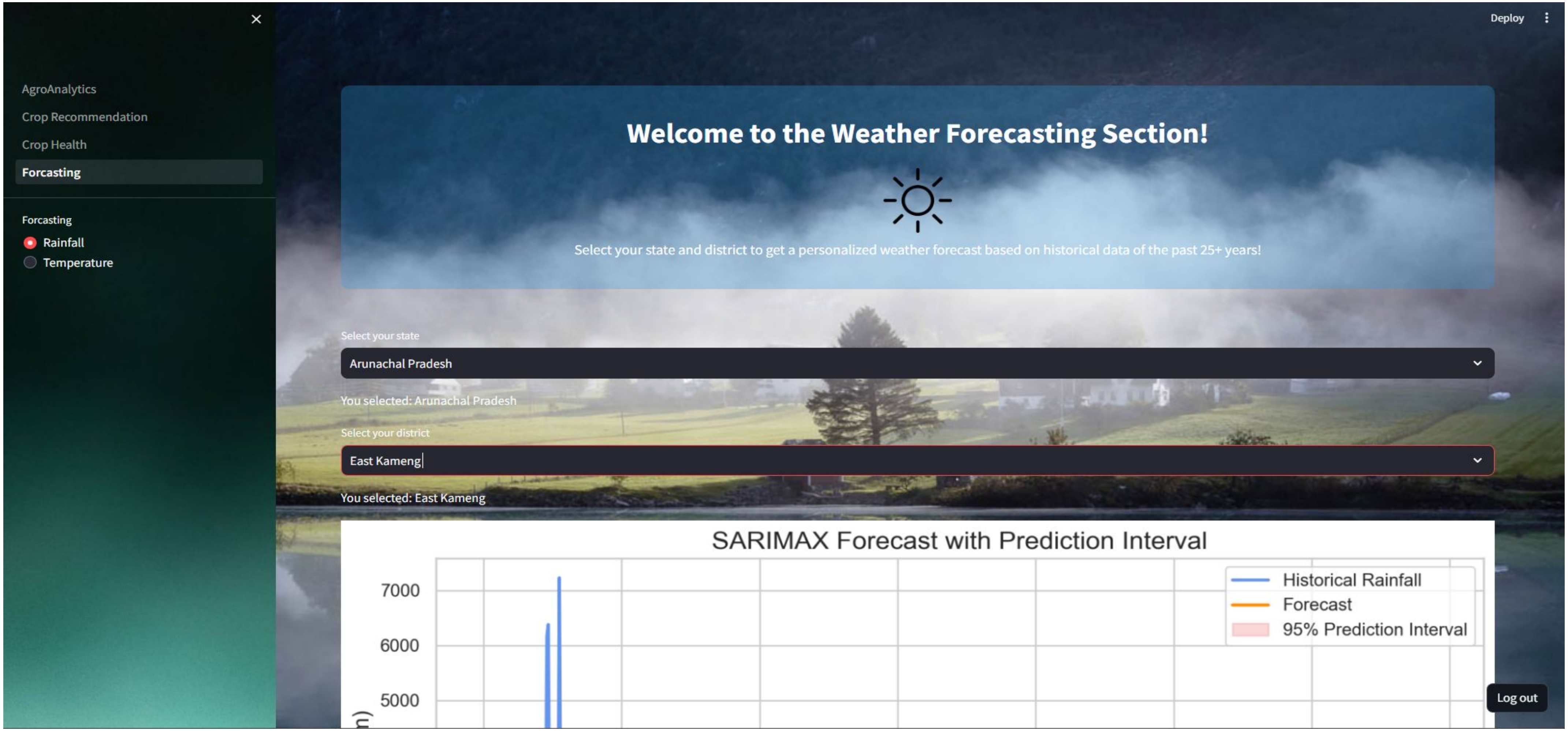


CORPORATE GURUKUL

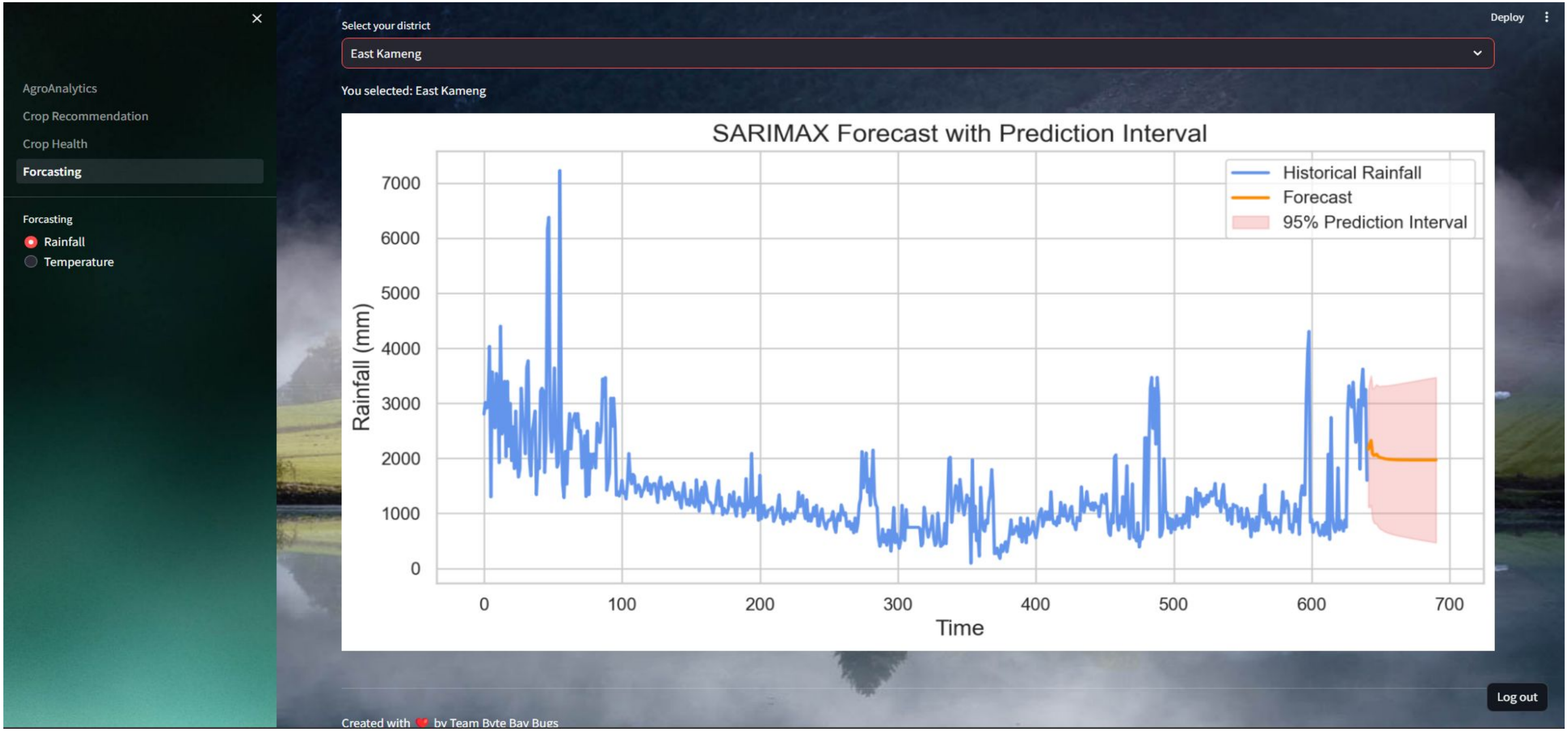
Crop Health Analysis Page



Weather Forecasting Page (Rainfall)



Weather Forecasting Page (Rainfall)



CORPORATE GURUKUL

Demo

CORPORATE GURUKUL

Challenges faced

CORPORATE
GURUKUL

Challenges faced

- **Data Accuracy,Availability and Quality:**
  - Precision crop recommendations rely on accurate and up-to-date data, including soil composition, weather patterns, and crop health information.
  - Obtaining high-quality data for leaf health assessment, including accurate images and relevant environmental data.
- **Computational Resources:**
  - Running computationally intensive models for long-term weather prediction requires significant computing resources.
- **Algorithm Complexity:**
  - Developing precise algorithms for crop recommendation that consider various factors such as soil nutrient levels, weather conditions, and historical data can be complex.

Future Enhancements

Future Enhancements

- ➔ Hosting it on a cloud based platform (like AWS) in order to save computational power, enhance collaborative workspace environment and maintain database integrity.
- ➔ Considering more class labels for crop recommendation and crop health analytics like using multiple factors that make up the composition of soil and affect its quality as well as the plants growth rate in our dataset
- ➔ Making it an open source framework for the research community to interact with on platforms like github and kaggle.
- ➔ Maintaining the privacy and security of our users while simultaneously providing them with our services by incorporating cybersecurity measures.
- ➔ Scaling our project to expand its reach to all those in need of this platform, worldwide.
- ➔ Making it more adaptable, scalable, computational and most importantly, accessible, to all our users.

CORPORATE GURUKUL

Our Learnings

CORPORATE GURUKUL

Our Learnings

- Implementation of Advanced Concepts of Machine Learning, Deep Learning and Computer Vision.
- Throughout the front-end development phase, challenges arose during the integration of back-end and front-end components. However, this experience provided valuable learning opportunities for our team.
- Throughout this brief two-week project, we gained a profound understanding of the significance of teamwork while working on a strict time crunch. Collaborating closely, we leveraged each other's strengths, that significantly contributed to our collective knowledge and skills.
- The process of training models and algorithms with our dataset presented a formidable challenge, offering a profound learning experience. This endeavor provided invaluable insights as we played around with Deep Learning models like OpenCV, SARIMAX, Random Forest, XGBoost, KNN Classifier and Naive Bayes.

CORPORATE GURUKUL

Conclusion

CORPORATE
GURUKUL

Conclusion

➔ This project integrating YOLOv3, XGBoost, and various machine learning algorithms holds promise for advancing precision agriculture. We've explored real-time object detection, and diverse algorithms for crop recommendations, as well as diverse models for leaf health assessment, and weather prediction.

➔ This underscores the transformative potential of technology in revolutionizing agricultural practices for enhanced precision and practicality.

CORPORATE
GURUKUL

BIBLIOGRAPHY

- Research Paper: PlantDoc: A Dataset for Visual Plant Disease Detection
- Trusted Friend, chatGPT
- Kaggle Dataset
- Roboflow dataset
- OpenCV Documentation
- Plotly Documentation- I II
- KNN Documentation, XGBoost Documentation, YT Resource 1 2

CORPORATE GURUKUL